

GARDEN RIVER, AB, Canada

WMO: 712530

Lat: 58.7108N

Lon: 113.8664W

Elev: 241

StdP: 98.47

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -39.8	(c) -36.7	(d) -44.1	(e) 0.1	(f) -40.0	(g) -40.9	(h) 0.1	(i) -37.2	(j) 5.1	(k) -10.6	(l) 4.4	(m) -12.2	(n) 0.2	(o) 230	(p) 0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	29.1	19.0	27.3	18.0	25.5	17.2	20.5	26.7	19.5	25.1	18.4	23.4	1.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.7	22.9	17.4	12.8	21.7	16.4	12.0	20.7	60.2	27.0	56.5	25.3	52.9	23.5	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 5.1	(b) 4.4	(c) 3.8	(e) -41.8	(f) 32.5	(g) 2.6	(h) 1.6	(i) -43.7	(j) 33.6	(k) -45.2	(l) 34.5	(m) -46.6	(n) 35.4	(o) -48.5	(p) 36.6		
(a) 5.1	(b) 4.4	(c) 3.8	(e) -42.6	(f) 22.4	(g) 2.0	(h) 1.8	(i) -44.1	(j) 23.6	(k) -45.3	(l) 24.6	(m) -46.4	(n) 25.6	(o) -47.9	(p) 26.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.3	-20.5	-16.3	-9.3	1.8	9.4	15.0	17.4	15.0	9.5	1.3	-10.1	-18.2		
	DBStd	14.68	8.79	7.44	8.60	5.98	4.84	3.48	2.91	3.40	4.03	4.45	6.68	8.33		
	HDD10.0	4407	944	736	597	250	69	4	0	3	57	271	602	874		
	HDD18.3	6852	1202	969	855	497	279	107	52	111	266	528	852	1133		
	CDD10.0	637	0	0	0	3	49	155	229	159	41	1	0	0		
	CDD18.3	41	0	0	0	0	1	8	23	8	1	0	0	0		
	CDH23.3	929	0	0	0	3	83	226	387	203	26	1	0	0		
	CDH26.7	226	0	0	0	0	17	49	106	49	4	0	0	0		
	WSAvg	1.4	1.0	1.2	1.6	2.0	2.0	1.6	1.3	1.3	1.3	1.5	1.2	1.0		
	PrecAvg	416	18	16	18	17	35	64	79	57	44	27	22	20		
(15) Precipitation	PrecMax	549	35	35	30	32	57	156	131	93	101	49	39	36		
	PrecMin	345	5	4	6	7	11	19	50	31	16	6	12	7		
	PrecStd	51	7	6	7	6	11	24	21	16	17	11	7	7		
	PrecStd	51	7	6	7	6	11	24	21	16	17	11	7	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	5.0	13.2	21.7	28.7	30.6	32.1	31.1	26.6	17.4	6.7	2.9		
		MCWB	1.3	1.5	5.7	10.9	15.5	18.5	19.9	21.0	17.8	9.9	3.3	1.6		
	2%	DB	-1.4	0.7	9.5	17.7	25.6	28.0	29.3	27.8	22.8	13.6	3.0	-1.1		
		MCWB	-2.0	-1.2	3.8	8.6	13.6	17.6	19.5	19.6	15.7	8.0	1.0	-1.9		
	5%	DB	-4.9	-2.3	6.6	14.9	22.8	26.0	27.5	25.5	20.0	10.7	0.6	-4.4		
		MCWB	-5.2	-3.7	2.1	7.1	12.5	16.5	18.8	18.3	14.0	6.8	-0.1	-4.7		
	10%	DB	-8.3	-5.4	3.3	12.4	19.9	23.9	25.4	23.4	17.5	8.2	-1.0	-6.9		
		MCWB	-8.7	-6.1	0.2	5.9	10.7	15.8	18.2	17.2	12.6	5.4	-1.5	-7.1		
	0.4%	WB	1.1	1.6	6.0	11.1	16.9	20.4	22.2	21.8	18.6	11.5	3.4	2.0		
		MCDB	2.5	4.5	12.8	19.9	25.5	27.5	27.9	28.1	25.5	16.1	5.8	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-2.4	-1.2	4.1	9.1	14.6	18.9	20.8	20.2	16.5	8.6	1.0	-1.5		
		MCDB	-1.7	0.4	9.0	16.3	22.9	24.8	27.0	26.0	21.5	12.7	2.0	-0.8		
	5%	WB	-5.9	-3.6	2.3	7.5	12.7	17.7	19.9	19.0	14.8	7.2	-0.3	-4.3		
		MCDB	-5.5	-2.5	6.2	13.8	20.5	23.2	25.7	23.8	18.7	10.6	0.4	-3.9		
	10%	WB	-9.6	-6.1	0.3	6.1	11.3	16.5	18.9	17.6	13.3	5.7	-1.9	-7.2		
		MCDB	-9.3	-5.2	3.3	11.9	18.3	22.0	24.1	21.9	16.7	8.3	-1.3	-6.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	13.6	15.6	14.7	15.3	15.1	14.3	14.4	13.5	9.9	8.6	9.4		
		MCDBR	12.8	16.6	17.9	19.2	21.3	19.4	18.3	18.2	18.4	15.0	9.6	11.1		
		MCWBR	12.7	14.3	12.3	11.0	10.0	9.4	9.3	10.1	10.7	9.8	8.0	10.9		
		MCDBR	13.2	16.3	17.2	18.0	18.2	16.1	16.5	16.7	16.6	14.5	9.0	11.7		
	5% WB	MCWBR	12.3	14.1	12.0	10.7	9.1	8.7	9.3	10.2	10.5	9.5	7.5	10.9		
		MCWBR	12.3	14.1	12.0	10.7	9.1	8.7	9.3	10.2	10.5	9.5	7.5	10.9		
(40) Clear-Sky Solar Irradiance	taub		0.217	0.242	0.270	0.329	0.343	0.368	0.387	0.350	0.316	0.286	0.232	0.197		
	taud		2.224	2.319	2.306	2.336	2.428	2.376	2.364	2.463	2.542	2.558	2.336	2.117		
	Ebn at Noon		654	819	886	878	889	867	839	851	832	764	636	540		
	Edh at Noon		42	65	89	104	103	110	109	92	72	53	38	32		
(44) All-Sky Solar Radiation	RadAvg		0.54	1.49	3.18	4.88	5.71	5.89	5.51	4.49	2.98	1.45	0.58	0.32		
	RadStd		0.04	0.08	0.17	0.28	0.47	0.43	0.28	0.24	0.22	0.17	0.07	0.03		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page